Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The incentre of a triangle with vertices $(7, 1), (-1, 5)$ and $(3 + 2\sqrt{3}, 3 + 4\sqrt{3})$ is

(A) $\left(3 + \frac{2}{\sqrt{3}}, 3 + \frac{4}{\sqrt{3}}\right)$

(B) $\left(1 + \frac{2}{3\sqrt{3}}, 1 + \frac{4}{3\sqrt{3}}\right)$

(C) $(7, 1)$

(D) None of these

Q.2

What is the radius of the circle passing through the point $(2, 4)$ and having centre at the intersection of the lines $x - y = 4$ and $2x + 3y + 7 = 0$?

(A) 3 units

(B) 5 units

(C) $3\sqrt{3}$ units

(D) $5\sqrt{2}$ units

Q.3

A line L intersects the three sides BC, CA and AB of a $\triangle ABC$ at P, Q and R respectively. Then, $\frac{BP}{PC} \cdot \frac{CQ}{QA} \cdot \frac{AR}{RB}$ is equal to

(A) 1

(B) 0

(C) -1

(D) None of these

Q.4

A regular polygon with equal sides has 9 diagonals. Two of the vertices are at $\mathrm{A}(-1, 0)$ and $\mathrm{B}(1, 0)$. Possible areas of polygon is

(A) $\frac{3\sqrt{3}}{2}, 2\sqrt{3}, 6\sqrt{3}$

(B) $2\sqrt{3}, 3\sqrt{3}, 6\sqrt{3}$

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(C) $9\sqrt{3}, 6\sqrt{3}, 2\sqrt{3}$

(D) $\frac{3\sqrt{3}}{2}, 3\sqrt{3}, 6\sqrt{3}$

Q.5

The area of the figure formed by the lines $ax + by + c = 0, ax - by + c = 0, ax + by - c = 0$ and $ax - by - c = 0$ is

(A) $\frac{c^2}{ab}$

(B) $\frac{2c^2}{ab}$

(C) $\frac{c^2}{2ab}$

(D) $\frac{c^2}{4ab}$

Q.6

If the point $P(x, y)$ is equidistant from the points $A(a + b, b - a)$ and $B(a - b, a + b)$, then

(A) $ax = by$

(B) $bx = ay$ and P can be (a, b)

(C) $x^2 - y^2 = 2(ax + by)$

(D) None of the above

Q.7

Let $A\left(\alpha, \frac{1}{\alpha}\right), B\left(\beta, \frac{1}{\beta}\right), C\left(\gamma, \frac{1}{\gamma}\right)$ be the vertices of a $\triangle ABC$, where α, β are the roots of the equation $x^2 - 6p_1x + 2 = 0, \beta, \gamma$ are the roots of the equation $x^2 - 6p_2x + 3 = 0$ and γ, α are the roots of the equation $x^2 - 6p_3x + 6 = 0, p_1, p_2, p_3$ being positive. Then, the coordinates of the centroid of $\triangle ABC$ is

(A) $\left(1, \frac{11}{18}\right)$

(B) $\left(0, \frac{11}{8}\right)$

(C) $\left(2, \frac{11}{18}\right)$

(D) None of these

Q.8

The line $x + y = a$ meets the axes of x and y at A and B respectively. A $\triangle AMN$ is inscribed in the $\Delta OAB, O$ being the origin, with right angle at N. M and N lie respectively on OB and AB. If the area of the $\triangle AMN$ is $\frac{3}{8}$ of the area of the ΔOAB, then $\frac{AN}{BN}$ is equal to

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(A) $\frac{1}{3}$

(B) $\frac{1}{3}, 3$

(C) $\frac{2}{3}, 3$

(D) 3

Q.9

If the line segment joining the points $\mathrm{A(a, b)}$ and $\mathrm{B(c, d)}$ subtends an angle θ at the origin, then $\cos\theta =$

(A) $\frac{ac+bd}{\sqrt{(a^2+b^2)(c^2+d^2)}}$

(B) $\frac{ab+cd}{\sqrt{(a^2+b^2)(c^2+d^2)}}$

(C) $\frac{ad+bc}{\sqrt{(a^2+b^2)(c^2+d^2)}}$

(D) None of these

Q.10

What is the equation of the line through $(1, 2)$ so that the segment of the line intercepted between the axes is bisected at this point?

(A) $2x - y = 4$

(B) $2x - y + 4 = 0$

(C) $2x + y = 4$

(D) $2x + y + 4 = 0$

Q.11

Through the point $P(\alpha, \beta)$, where $\alpha\beta > 0$, the straight line $\frac{x}{a} + \frac{y}{b} = 1$ is drawn so as to form with axes a triangle of area S. If $ab > 0$, then least value of S is

(A) $\alpha\beta$

(B) $2\alpha\beta$

(C) $3\alpha\beta$

(D) None of these

Q.12

The middle point of the segment of the straight line joining the points (p, q) and $(q, -p)$ is $(r/2, s/2)$. What is the length of the segment?

(A) $\left[\left(s^2 + r^2\right)^{1/2}\right]/2$

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(B) $\left[\left(s^2+r^2\right)^{1/2}\right]/4$

(C) $\left(s^2+r^2\right)^{1/2}$

(D) $s+r$

Q.13

If the sum of the squares of the distances of the point (x, y) from the points $(a, 0)$ and $(-a, 0)$ is $2b^2$, then which one of the following is correct

(A) $x^2+a^2=b^2+y^2$

(B) $x^2+a^2=2b^2-y^2$

(C) $x^2-a^2=b^2+y^2$

(D) $x^2+a^2=b^2-y^2$

Q.14

A rectangle $ABCD$, where $A(0,0),\quad B(4,0)\ C(4,2), D(0,2)$, undergoes the following transformations successively:

$f_1(x, y) \rightarrow (y, x)$

ii. $f_2(x, y) \rightarrow (x+3y, y)$

iii. $f_3(x, y) \rightarrow ((x-y)/2, (x+y)/2)$

The final figure will be

(A) a square

(B) a rhombus

(C) a rectangle

(D) a parallelogram

Q.15

A straight line through the origin O meets the parallel lines $4\mathrm{x}+2\mathrm{y}=9$ and $2\mathrm{x}+y+6=0$ at points P and Q respectively, then point O divides the segment PQ in the ratio:

(A) $1:2$

(B) $3:4$

(C) $2:1$

(D) $4:3$

Q.16

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For $a > b > c > 0$, the distance between $(1, 1)$ and the point of intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then

(A) $a + b - c > 0$

(B) $a - b + c < 0$

(C) $a - b + c > 0$

(D) $a + b - c < 0$

Q.17

The straight lines $7x - 2y + 10 = 0$ and $7x + 2y - 10 = 0$ forms an isosceles triangle with the line $y = 2$. Area of this triangle is equal to

(A) $\frac{15}{7}$ sq. units

(B) $\frac{10}{7}$ sq. units

(C) $\frac{18}{7}$ sq. units

(D) none of these

Q.18

As system of line is given as $y = m_i x + c_i$, where m_i can take any value out of $0, 1, -1$ and when m_j is positive then c_i can be 1 or -1 when m_i equal 0, c_i can be 0 or 1 and when m_i equals -1, c_i can take 0 or 2 . Then the area enclosed by all these straight line is

(A) $\frac{3}{\sqrt{2}}(\sqrt{2} - 1)$

(B) $\frac{3}{\sqrt{2}}$

(C) $\frac{3}{2}$

(D) none of these

Q.19

If the point $(x_1 + t(x_2 - x_1), y_1 + t(y_2 - y_1))$ divides the join of (x_1, y_1) and (x_2, y_2) internally, then

(A) $r < 0$

(B) $0 < t < 1$

(C) $t > 1$

(D) $t = 1$

Q.20

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The vertices of a triangle are $(pq, 1/(pq)), (qr, 1/qr))$ and $(rq, 1/(m))$ where p, q, r are the roots of the equation $y^3 - 3y^2 + 6y + 1 = 0$. The coordinates of its centroid are

(A) $(1, 2)$

(B) $(2, -1)$

(C) $(1, -1)$

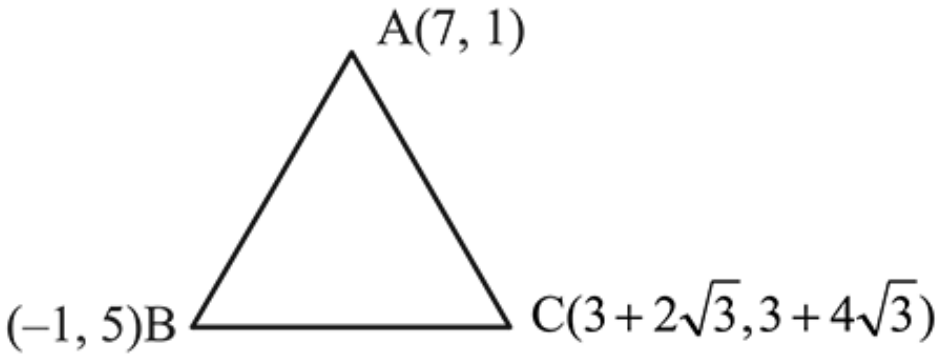
(D) $(2, 3)$

Answers & Solutions

Q.1 Answer:

$\left(3 + \frac{2}{\sqrt{3}}, 3 + \frac{4}{\sqrt{3}}\right)$

Solution:



$\because AB = BC = CA = 4\sqrt{5},$

i.e., given triangle is equilateral. (Incentre of a triangle are same as the centroid when triangle is equilateral)

Hence, incentre

$= \left(\frac{7-1+3+2\sqrt{3}}{3}, \frac{1+5+3+4\sqrt{3}}{3}\right)$

$= \left(3 + \frac{2}{\sqrt{3}}, 3 + \frac{4}{\sqrt{3}}\right)$

Q.2 Answer:

$5\sqrt{2}$ units

Solution:

We have $x - y = 4 \& 2x + 3y + 7 = 0$

On solving, we get, $x = 1 \& y = -3$

(these are coordinates of centre of the circle) $\Rightarrow$ **radius** $= \sqrt{(2-1)^2 + (4+3)^2} = 5\sqrt{2}$

Q.3 Answer:

-1

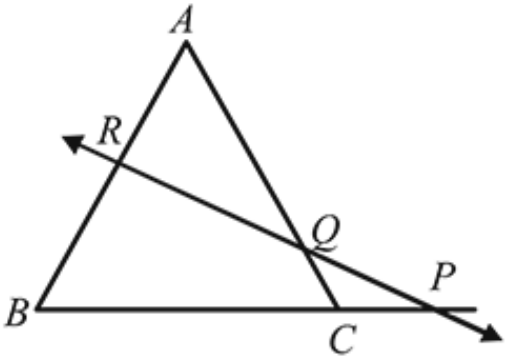
Solution:

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Let $A\left(x_1, y_1\right), B\left(x_2, y_2\right)$ and $C\left(x_3, y_3\right)$ be the vertices of $\triangle ABC$ and let $lx+my+n=0$ be the equation of the line. If P divides BC in the ratio $\lambda: 1$, then the coordinates of P are

$l\left(\frac{\lambda x_3+x_2}{\lambda+1}\right)+m\left(\frac{\lambda y_3+y_2}{\lambda+1}\right)+n=0$

$\Rightarrow -\frac{lx_2+my_2+n}{lx_3+my_3+n}=\lambda=\frac{BP}{PC}$



Similarly, we obtain

$\frac{CQ}{QA}=-\frac{lx_3+my_3+n}{lx_1+my_1+n}$

and $\frac{AR}{RB}=-\frac{lx_1+my_1+n}{lx_2+my_2+n}$

On multiplying Eqs. (i), (ii) and (iii), we get

$\frac{BP}{PC}\cdot\frac{CQ}{QA}\cdot\frac{AR}{RB}=-1$

Q.4 Answer:

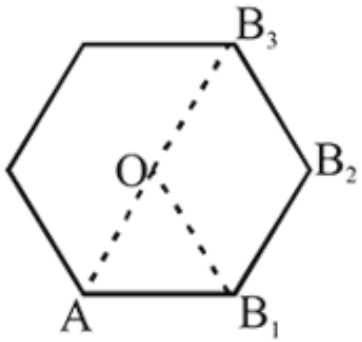
$\frac{3\sqrt{3}}{2}, 2\sqrt{3}, 6\sqrt{3}$

Solution:

If polygon has n **sides, then the number of diagonals**

$=\frac{\mathrm{n}(\mathrm{n}-3)}{2}=9$

$\mathrm{n}=6$

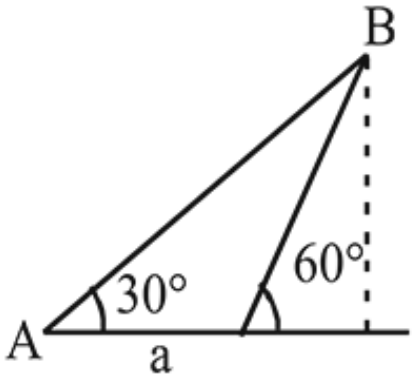


Now A **and** B **can be adjacent vertices attenate vertices or opposite vertices If** A **and** B **are adjacent then side** $AB=2$**, then area** $=6\times\Delta\mathrm{OAB}$

i.e area $=6\times\frac{\sqrt{3}}{4}\times(2)^2=6\sqrt{3}$

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If A and B are alternate, then $2\cos 30^\circ = a + a\cos 60^\circ$



$\therefore$ **side** $a = \frac{2}{\sqrt{3}}$

$\therefore$ **area** $= 6 \times \frac{\sqrt{3}}{4}\left(\frac{2}{\sqrt{3}}\right)^2 = 2\sqrt{3}$

Finally ifA and B are opposite vertices then side

$a = \frac{1}{2}AB = 1$

Then area $= 6 \times \frac{\sqrt{3}}{4}(1)^2 = \frac{3\sqrt{3}}{2}$

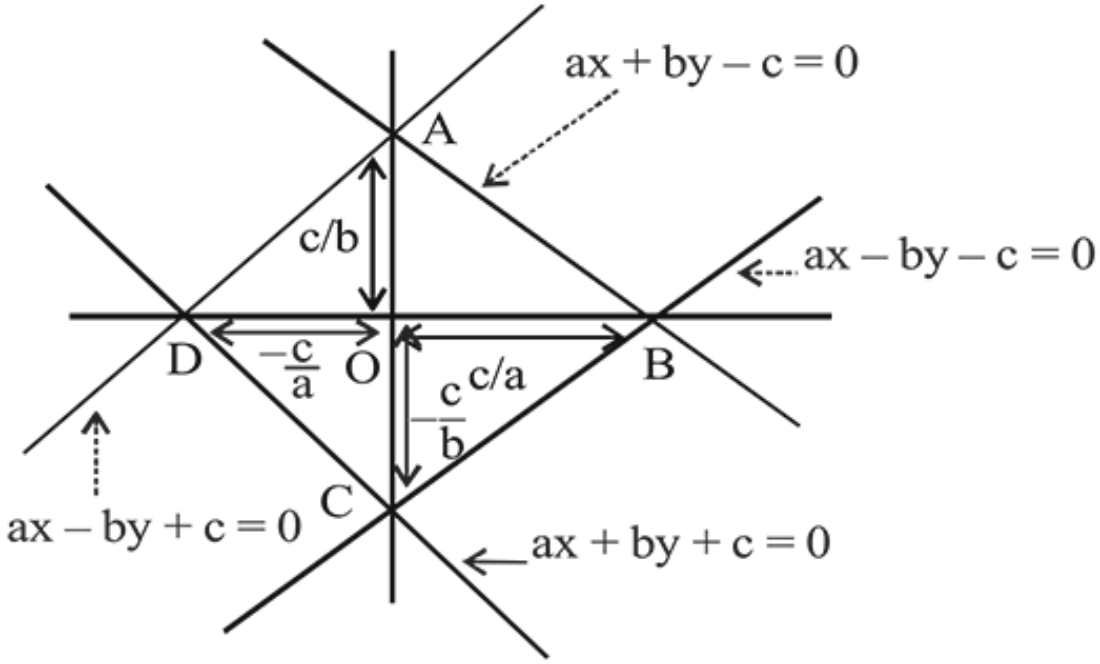
Q.5 Answer:

$\frac{2c^2}{ab}$

Solution:

Area of triangle

$\Delta AOB = \frac{1}{2} \times \frac{c}{b} \times \frac{c}{a} = \frac{c^2}{2ab}$



Total area $= 4\times$ **area** ΔAOB

$= 4 \times \frac{c^2}{2ab} = \frac{2c^2}{ab}$

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Q.6 Answer:

$bx = ay$ and P can be (a, b)

Solution:

We have, $PA = PB \Rightarrow (PA)^2 = (PB)^2$

$\Rightarrow [x-(a+b)]^2 + [y-(b-a)]^2$

$= [x-(a-b)]^2 + [y-(a+b)]^2$

$\Rightarrow [(x-a)-b]^2 + [(y-b)+a]^2$

$= [(x-a)+b]^2 + [(y-b)-a]^2$

$\Rightarrow [(x-a)+b]^2 - [(x-a)-b]^2$

$= [(y-b)+a]^2 - [(y-b)-a]^2$

$\Rightarrow 4b(x-a) = 4a(y-b) \Rightarrow bx = ay \ldots$ i)

Also, $P(a, b)$ satisfies the conditon (i), so that P can be (a, b).

Q.7 Answer:

$\left(2, \frac{11}{18}\right)$

Solution:

It is given that α, β are the roots of the equation $x^2 - 6p_1x + 2 = 0$

$\therefore \quad \alpha + \beta = 6\mathrm{p}_1, \alpha\beta = 2$ ...(i)

β, γ are the roots of the equation $x^2 - 6p_2x + 3 = 0$

$\therefore \quad \beta + \gamma = 6\mathrm{p}_2, \beta\gamma = 3$ ...(ii)

γ, α are the roots of the equation $x^2 - 6p_3x + 6 = 0 \therefore \quad \gamma + \alpha = 6\mathrm{p}_3, \gamma\alpha = 6$ ...(iii)

From Eqs. (i), (ii) and (iii), we get $\Rightarrow \quad \alpha\beta\gamma = 6 \quad [\because \alpha, \beta, \gamma > 0]$

Now, $\alpha\beta = 2$ and $\alpha\beta\gamma = 6$

$\Rightarrow \quad \gamma = 3$

$\beta\gamma = 3$ and $\alpha\beta\gamma = 6$

$\alpha = 3 \quad \alpha = 6\alpha\beta\gamma = 6$

$\Rightarrow \quad \beta = 1$

$\therefore \quad \alpha + \beta = 6p_1 \Rightarrow 3 = 6p_1$

$\Rightarrow \quad \mathrm{p}_1 = \frac{1}{2}$

$\beta + \gamma = 6p_2 \Rightarrow 4 = 6p_2$

$\Rightarrow \quad \mathrm{p}_2 = \frac{2}{3}$

and $\gamma + \alpha = 6p_3 \Rightarrow 5 = 6p_3$

$\Rightarrow \quad \mathrm{p}_3 = \frac{5}{6}$

The coordinates of the centroid of triangle are

$\left(\frac{\alpha+\beta+\gamma}{3}, \frac{1}{3}\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right)\right)$ or

$\left(\frac{6}{3}, \frac{1}{3}\left(\frac{1}{2} + 1 + \frac{1}{3}\right)\right)$ or $\left(2, \frac{11}{18}\right)$

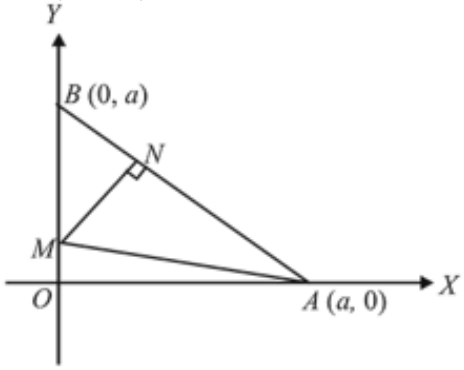
Q.8 Answer:

3

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Solution:

Let $\frac{AN}{BN} = \lambda$. Then, the coordinates of N are $\left(\frac{a}{1+\lambda}, \frac{\lambda a}{1+\lambda}\right)$



Where $(a, 0)$ and $(0, a)$ are the coordinates of A and B respectively. Now, equation of MN perpendicular to AB is

$y - \frac{\lambda a}{1+\lambda} = x - \frac{a}{1+\lambda}$

$\Rightarrow x - y = \frac{1-\lambda}{1+\lambda} a$

So, the coordinates of M are $\left(0, \frac{\lambda-1}{\lambda+1} a\right)$

Therefore, area of the $\triangle AMN$ is

$= \frac{1}{2} \left| \left[a\left(\frac{-a}{\lambda+1}\right) + \frac{1-\lambda}{(1+\lambda)^2} a^2 \right] \right.$

$= \frac{\lambda a^2}{(1+\lambda)^2}$

Also, area of $\Delta OAB = \frac{a^2}{2}$

So, that according to the given condition

$\frac{\lambda a^2}{(1+\lambda)^2} = \frac{3}{8} \cdot \frac{1}{2} a^2$

$\Rightarrow 3\lambda^2 - 10\lambda + 3 = 0$

$\Rightarrow \lambda = 3$ **or** $\lambda = \frac{1}{3}$

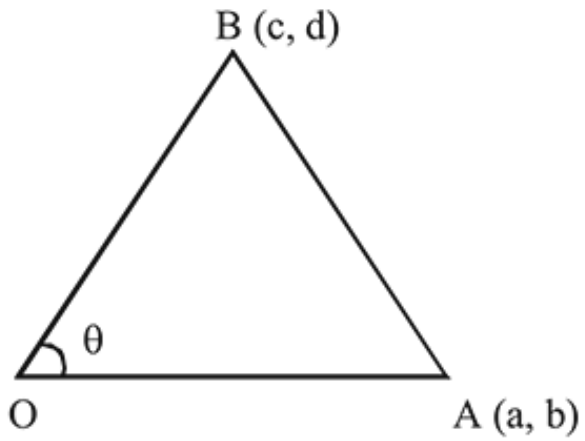
For $\lambda = \frac{1}{3}$, M lies outside the segment OB and hence the required value is 3

Q.9 Answer:

$\frac{ac+bd}{\sqrt{(a^2+b^2)(c^2+d^2)}}$

Solution:

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b) Let the origin be O. **So** $\mathrm{O} = (0, 0)$ **Now** $\mathrm{AB}^2 = (\mathrm{a} - \mathrm{c})^2 + (\mathrm{b} - \mathrm{d})^2$

$OA^2 = (a - 0)^2 + (b - 0)^2 = a^2 + b^2$

and $OB^2 = (c - 0)^2 + (d - 0)^2 = c^2 + d^2$

Now from the $\triangle \mathrm{AOB}$ **:**

$\cos\theta = \frac{\mathrm{OA}^2 + \mathrm{OB}^2 - \mathrm{AB}^2}{2\mathrm{OA} \cdot \mathrm{OB}}$

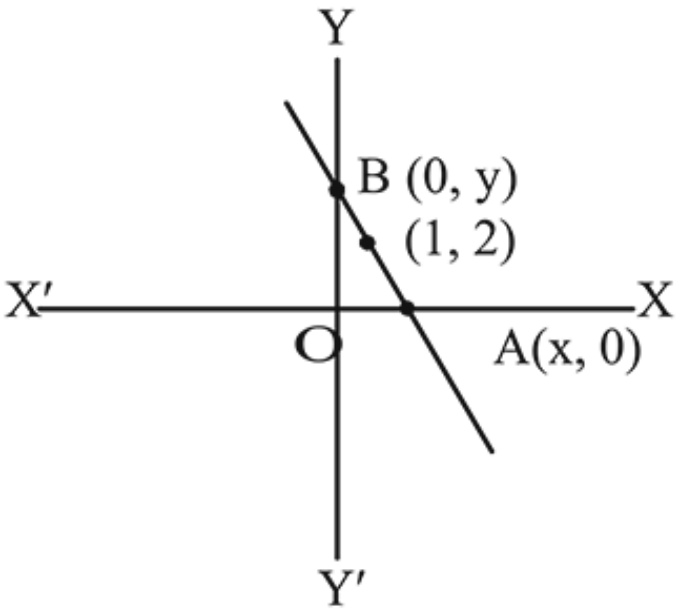
$= \frac{a^2+b^2+c^2+d^2-\left\{(a-c)^2+(b-d)^2\right\}}{2\sqrt{a^2+b^2}\sqrt{c^2+d^2}}$

$= \frac{2(ac+bd)}{2\sqrt{(a^2+b^2)(c^2+d^2)}} = \frac{ac+bd}{\sqrt{(a^2+b^2)(c^2+d^2)}}$

Q.10 Answer:

$2x + y = 4$

Solution:



$\frac{0+\mathrm{x}}{2} = 1; \frac{0+\mathrm{y}}{2} = 2$

$x = 2; y = 4$

Equation of line passing through $(2, 0)$ **and** $(0, 4)$ $y - 0 = \frac{4-0}{0-2}(x - 2)$

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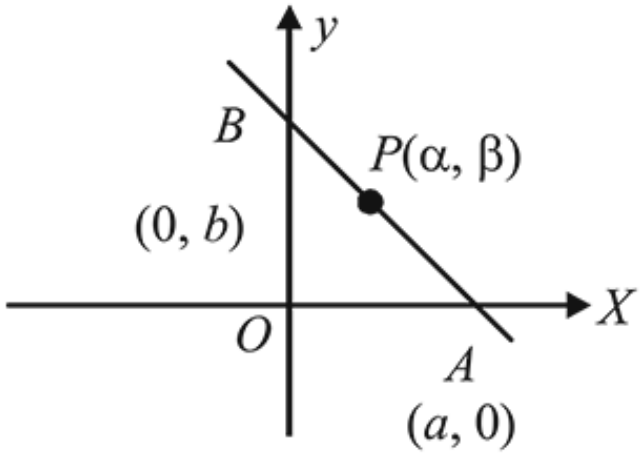
$y = -2x + 4$

$2x + y = 4$

Q.11 Answer:

$2\alpha\beta$

Solution:



Area of $\Delta OAB = S = \frac{1}{2}ab$

Equation of AB **is** $\frac{x}{a} + \frac{y}{b} = 1$

Putting (α, β)**, we get** $\frac{\alpha}{a} + \frac{\beta}{b} = 1$

$\Rightarrow \frac{\alpha}{a} + \frac{a\beta}{2S} = 1$ **[using (i)]**

$\Rightarrow a^2\beta - 2aS + 2\alpha S = 0 \quad \therefore a \in R \Rightarrow D \geq 0$

$4S^2 - 8\alpha\beta S \geq 0$

$\Rightarrow S \geq 2\alpha\beta$**. Least value of** $S = 2\alpha\beta$**.**

Q.12 Answer:

$\left(s^2 + r^2\right)^{1/2}$

Solution:

Two joining points are (p, q) **and** $(q, -p)$ **Mid point of** (p, q) **and** $(q, -p)$ **is** $\left(\frac{p+q}{2}, \frac{q-p}{2}\right)$

But it is given that the mid-point is $\left(\frac{r}{2}, \frac{s}{2}\right)$**.**

$\therefore \frac{p+q}{2} = \frac{r}{2}$ **and** $\frac{q-p}{2} = \frac{s}{2}$

$\Rightarrow p + q = r$ **and** $q - p = s$

Now, length of segment $= \sqrt{(p-q)^2 + (q+p)^2}$

(by distance formula)

$= \sqrt{s^2 + r^2}$

Q.13 Answer:

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$x^2 + a^2 = b^2 - y^2$

Solution:

Let $\mathrm{P}(x, y)$ **be a point and** $\mathrm{A} = (a, 0), \mathrm{B} = (-a, 0)$.

Now, $\mathrm{PA}^2 = (x - a)^2 + y^2$

$\mathrm{PB}^2 = (x + a)^2 + y^2$

Since the sum of the distances of the point $\mathrm{P}(x, y)$ **from the points** $\mathrm{A}(a, 0)$ **and** $\mathrm{B}(-a, 0)$ **is** $2b^2$.

$\therefore \quad \mathrm{PA}^2 + \mathrm{PB}^2 = 2b^2$

$(x - a)^2 + (y - 0)^2 + (x + a)^2 + (y - 0)^2 = 2b^2$

$\Rightarrow x^2 + a^2 - 2ax + y^2 + x^2 + a^2 + 2ax + y^2 = 2b^2$

$\Rightarrow x^2 + a^2 + y^2 = b^2$

$\Rightarrow x^2 + a^2 = b^2 - y^2$

Q.14 Answer:

a parallelogram

Solution:

Clearly, A **will remain as** $(0, 0)$; f_1 **will make** b **as** $(0, 4)$, f_2 **will makeit** $(12, 4)$, **and** f_3 **will make it** $(4$

8); f_1 **will make** C **as** $(2, 4)$, f_2 **will make it** $(14, 4)$, **and** f_3 **will make it** $(5, 9)$. **Finally,** f_1 **will make** D **as** $(2, 0)$, f_2 **will make it** $(2, 0)$, **and** f_3 **will make it** $(1, 1)$. **So, we finally get**

$A(0, 0), B(4, 8), C(5, 9)$, **and**

$D(1, 1)$. **Hence,**

$m_{AB} = \frac{8}{4}, m_{BC} = \frac{9-8}{5-4} = 1, m_{CD} = \frac{9-1}{5-1} = \frac{8}{4},$

$m_{AD} = 1, m_{AC} = \frac{9}{5}, m_{BD} = \frac{8-1}{4-1} = \frac{7}{3}$

Hence, the final figure will be a parallelogram.

Q.15 Answer:

$3 : 4$

Solution:

Let, equation of line passing through origin is $y = mx - (1)$

$4x + 2y = 9 - (2)$

$2x + y = -6 - (3)$

From intersection of line 1 with line 2

$\mathrm{x}_1 = \frac{9}{4+2\,\mathrm{m}} \& \mathrm{y}_1 = \frac{9\,\mathrm{m}}{4+2\,\mathrm{m}} \mathrm{P}\left(\frac{9}{4+2\,\mathrm{m}}, \frac{9\,\mathrm{m}}{4+2\,\mathrm{m}}\right)$

From intersection of line 1 with line 3

$\mathrm{x}_2 = \frac{-6}{2+\mathrm{m}} \& \mathrm{y}_2 = \frac{-6}{2+\mathrm{m}} \mathrm{Q}\left(\frac{-6}{2-\mathrm{m}}, \frac{-6\,\mathrm{m}}{2+\mathrm{m}}\right)$

Let o divides PQ in ration $1 : \mathrm{K}$

So, by formula

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$0=\frac{k\left(\frac{9}{4+2m}\right)-\frac{6}{2+m}}{k+1} \& 0=\frac{k\left(\frac{9m}{4+2m}\right)-\frac{6m}{2+1}}{k+1}$

$k=\frac{6\times 2}{9}=\frac{4}{3}$

$\therefore k=\frac{4}{3}$

$\therefore$ ration $\Rightarrow 1:k$

$1:\frac{4}{3}$

$\Rightarrow 3:4$

Let, equation of line passing through origin is $y=mx-(1)$

$4x+2y=9--(2)$

$2x+y=-6-(3)$

From intersection of line 1 with line 2

$x_1=\frac{9}{4+2m} \& y_1=\frac{9m}{4+2m} P\left(\frac{9}{4+2m},\frac{9m}{4+2m}\right)$

From intersection of line 1 with line 3

$x_2=\frac{-6}{2+m} \& y_2=\frac{-6}{2+m} Q\left(\frac{-6}{2-m},\frac{-6m}{2+m}\right)$

Let o divides PQ in ration $1:K$

So, by formula

$0=\frac{k\left(\frac{9}{4+2m}\right)-\frac{6}{2+m}}{k+1} \& 0=\frac{k\left(\frac{9m}{4+2m}\right)-\frac{6m}{2+1}}{k+1}$

$k=\frac{6\times 2}{9}=\frac{4}{3}$

$\therefore k=\frac{4}{3}$

$\therefore$ ration $\Rightarrow 1:k$

$1:\frac{4}{3}$

$\Rightarrow 3:4$

Q.16 Answer:

$a+b-c>0$

Solution:

The intersection point of two lines is $\left(\frac{-c}{a+b},\frac{-c}{a+b}\right)$

Distance between $(1,1)$ and $\left(\frac{-c}{a+b},\frac{-c}{a+b}\right)<2\sqrt{2}$

$\Rightarrow 2\left(1+\frac{c}{a+b}\right)^2<8$

$\Rightarrow \quad 1+\frac{c}{a+b}<2 \Rightarrow a+b-c>0$

Q.17 Answer:

$\frac{18}{7}$ sq. units

Solution:

lines $7x-2y+10=0, 7x+2y-10=0 \& y=2$ forms isosceles triangle

lines $7x-2y+10=0, \& y=2$ intersects in point $[-(6/7)\},2]$

lines $7x-2y+10=0, \& 7x+2y-10$ intersects in point $(0,5)$

lines $7x+2y-10=0, \& y=2$ intersects in point $[(6/7),2]$

$$\therefore \text{Area}=(1/2)\begin{vmatrix}1 & 0 & 5\\ 1 & (6/7) & 2\\ 1 & -(6/7) & 2\end{vmatrix}$$

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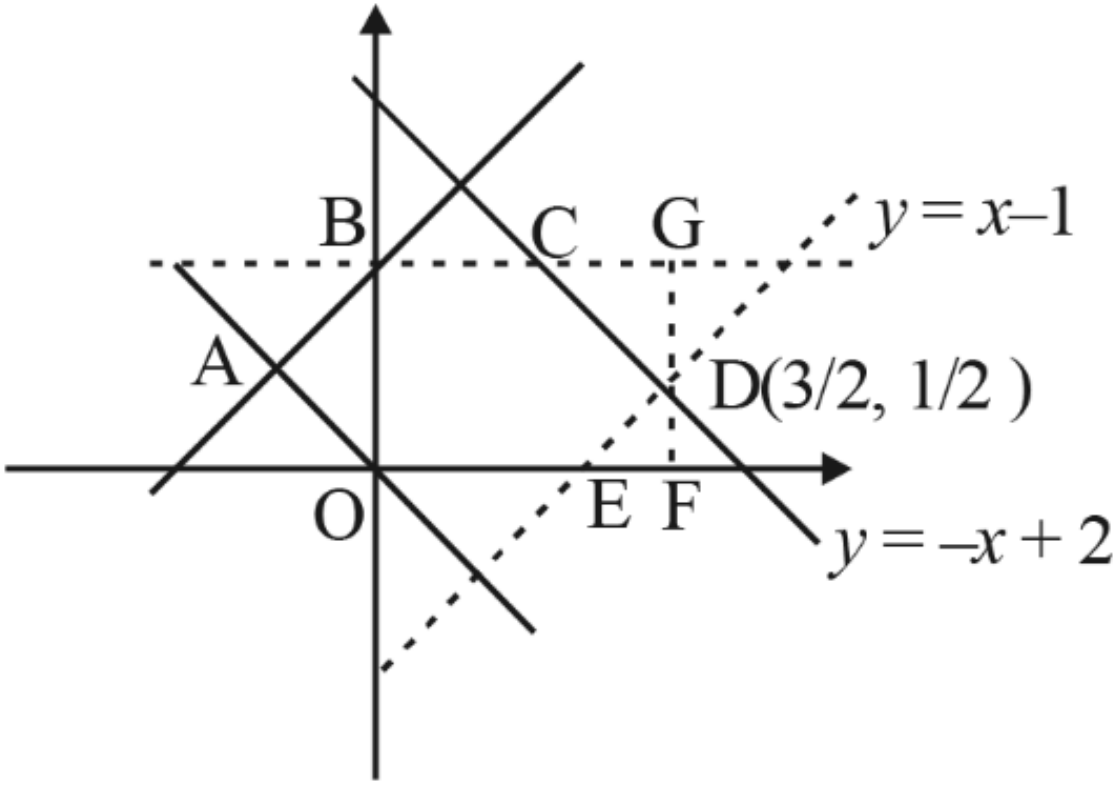
$= (1/2)|[\{(12/7) + (12/7)\} + 5\{-(6/7) - (6/7)\}]|$
$= (1/2)|[(24/7) - (60/7)]|$
$= (1/2) \times (36/7)$
$= (18/7)$ **sq. unit.**

Q.18 Answer:

$\frac{3}{2}$

Solution:

Lines are $y = 1, y = 0$
$y = -x, y = -x + 2$
$y = x + 1, y = x - 1$
Area of $OABCDE =$ **area of** $OBGF$
$= \frac{3}{2} \times 1 = \frac{3}{2}$



Q.19 Answer:

$0 < t < 1$

Solution:

$(x_1 + t(x_2 - x_1), y_1 + t(y_2 - y_1)) = (x_1(1 - t) + tx_2, y_1(1 - t) + ty_2)$ **is the point which divides** (x_1, y_1) **and** (x_2, y_2) **in the ratio** $t : 1 - t$
Since, the point divides internally
∴ Ratio must be positive.
∴ $\frac{t}{1-t} > 0$ **and** $t(1 - t) > 0$
Thus, $0 < t < 1$.

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Q.20 Answer:

$(2, -1)$

Solution:

p, q2r are the roots of equation $y^3 - 3y^2 + 6y + 1 = 0$. **so,** $\mathrm{p} + \mathrm{q} + \mathrm{r} = 3, \mathrm{pq} + \boldsymbol{q} + \mathrm{rp} = 6$ **and**

$\mathbf{pq} = -1$. **Now, the centroid of the triangle is**

$\left(\frac{pq+qr+m}{3}, \frac{\frac{1}{pq}+\frac{1}{qr}+\frac{1}{rp}}{3}\right)$

i.e., $\left(\frac{pq+qr+rp}{3}, \frac{p+q+r}{3pqr}\right) = \left(\frac{6}{3}, \frac{3}{-3}\right)$

or $(2, -1)$

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